

Proof: A Consistent Heuristic is Admissible

Problem Statement

A heuristic is consistent iff, for every node n and every successor n' of n generated by any action a ,

$$h(n) \leq c(n, a, n') + h(n')$$

Prove that a consistent heuristic is always admissible.

Definitions

Consistent Heuristic: A heuristic h is consistent if, for every node n and every successor n' of n generated by action a :

$$h(n) \leq c(n, a, n') + h(n') \quad (1)$$

Admissible Heuristic: A heuristic h is admissible if it never overestimates the true cost to reach the goal node. For every node n :

$$h(n) \leq h^*(n) \quad (2)$$

where $h^*(n)$ is the true minimum cost from n to the goal.

Proof

We prove this by mathematical induction on the number of nodes k on the shortest path to any goal from node n .

Base Case ($k = 1$):

Let n' be the goal node. Then n is 1 step away from the goal. By the definition of consistency:

$$h(n) \leq c(n, a, n') + h(n') \quad (3)$$

Since n' is the goal node, the cost to reach the goal from itself is zero, meaning $h(n') = 0$. Thus:

$$h(n) \leq c(n, a, n') \quad (4)$$

Because n' is the goal, the true optimal cost $h^*(n)$ is exactly the step cost $c(n, a, n')$. Therefore:

$$h(n) \leq h^*(n) \quad (5)$$

The base case holds.

Inductive Hypothesis:

Assume that for an arbitrary node n' which is k steps away from the goal on the optimal path, the heuristic is admissible. That is:

$$h(n') \leq h^*(n') \quad (6)$$

Inductive Step ($k + 1$ steps):

Consider a node n that is $k + 1$ steps from the goal. Let its successor on the optimal path to the goal be n' . Thus, n' is k steps from the goal.

By the property of consistency:

$$h(n) \leq c(n, a, n') + h(n') \quad (7)$$

Using our inductive hypothesis ($h(n') \leq h^*(n')$), we can substitute $h(n')$:

$$h(n) \leq c(n, a, n') + h^*(n') \quad (8)$$

Since n' lies on the shortest path from n to the goal, the exact cost from n to the goal, $h^*(n)$, is the sum of the step cost from n to n' and the exact cost from n' to the goal. Mathematically:

$$h^*(n) = c(n, a, n') + h^*(n') \quad (9)$$

Substituting this into our inequality yields:

$$h(n) \leq h^*(n) \quad (10)$$

This proves that $h(n)$ is also admissible for a node $k + 1$ steps from the goal.

Conclusion

By mathematical induction, $h(n) \leq h^*(n)$ for all nodes n . Therefore, any consistent heuristic is also an admissible heuristic.